BUSINESS METHOD FOR A PAYLOAD SIZE SURCHARGE IN A HIGH-BANDWIDTH DATA TRANSFER SERVICE

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Title: BUSINESS METHOD FOR A PAYLOAD SIZE SURCHARGE IN A HIGH-BANDWIDTH DATA TRANSFER SERVICE

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Abstract

A business method for applying a surcharge to a data transfer service based on the duration of the data transfer, where said duration is a function of the dataset's size. The method is specifically applicable to services utilizing ultra-high-bandwidth interfaces (e.g., ¿1 Terabit per second) where the data transfer time, while short, constitutes a non-negligible service time for a logistics operator (e.g., a delivery driver) who must wait for the transfer to complete. The method comprises: calculating the expected transfer duration based on the dataset size and interface speed, determining if the duration exceeds a base service time, and applying a tiered surcharge based on the excess time, thereby compensating for the operator's waiting time, which becomes the primary serial bottleneck in the process.

Background of the Invention

In traditional data transfer services, whether physical or digital, the actual time spent on the final data transfer is either negligible (for small files) or so long (for large files over slow connections) that it is not billed as a distinct "waiting time" fee. However, the advent of ultra-high-bandwidth interfaces, capable of transferring petabytes of data in minutes, creates a new economic scenario. A delivery driver for a physical data logistics service might arrive at a destination and have to wait for a multi-minute data transfer to complete. This waiting time is a direct cost to the service provider. A method is needed to fairly account for and bill for this "last-mile serial bottleneck" where the human operator's time, not the technology, is the limiting factor.

Summary of the Invention

The present invention is a business method for implementing a "payload size surcharge" in a high-speed data transfer service. The method addresses the economic bottleneck created when a human-operated part of a logistics chain must wait for an automated, high-speed, but time-consuming, data transfer to complete.

The method involves a pricing system for a data transfer service that includes a base fee for the service and a variable surcharge. The process is as follows: When a customer orders a data transfer, the size of the data payload is determined. The system calculates the expected transfer time by dividing the payload size by the known speed of the data transfer interface. This calculated time is compared to a pre-defined base service time (e.g., 30 seconds). If the transfer time exceeds this base, a surcharge is calculated based on the excess waiting time. The surcharge may be tiered, increasing for each additional minute or block of time the logistics operator must wait. This method allows a service provider to be compensated for the primary resource consumed during the final transfer: the operator's time.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental,

defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a flowchart illustrating the steps of calculating and applying the payload size surcharge.

Detailed Description of the Invention

Referring to **FIG. 1**, the method for applying a payload size surcharge is shown. The process begins when a data transfer service is requested (101). The size of the data payload is determined (102). The system then calculates the expected data transfer duration by dividing the payload size by the interface's transfer rate (103). A decision is then made based on whether this calculated duration exceeds a defined base service time (104). If it does not, only the base service fee is applied (105). If it does, a surcharge is calculated based on the excess time (106). This surcharge is then added to the base fee to generate the total service fee (107), which is then billed to the customer.

Claims

What is claimed is:

1. A method for billing for a data transfer service, the method comprising the steps of:
    - determining a size of a data payload to be transferred;
    - using a computer to calculate an expected transfer duration based on the data payload size and a pre-defined data transfer rate of a high-bandwidth interface;
    - comparing the expected transfer duration to a pre-defined base service time; and
    - applying a surcharge to a base fee for the service if the expected transfer duration exceeds the base service time.
2. The method of claim 1, wherein the surcharge is calculated based on the amount of time by which the expected transfer duration exceeds the base service time.

3. The method of claim 1, wherein the high-bandwidth interface has a data transfer rate exceeding 1 terabit per second.

4. The method of claim 1, wherein the service involves a logistics operator who must wait on-site for the duration of the data transfer.

5. The method of claim 4, wherein the surcharge is calculated to compensate for the waiting time of the logistics operator.

Drawings

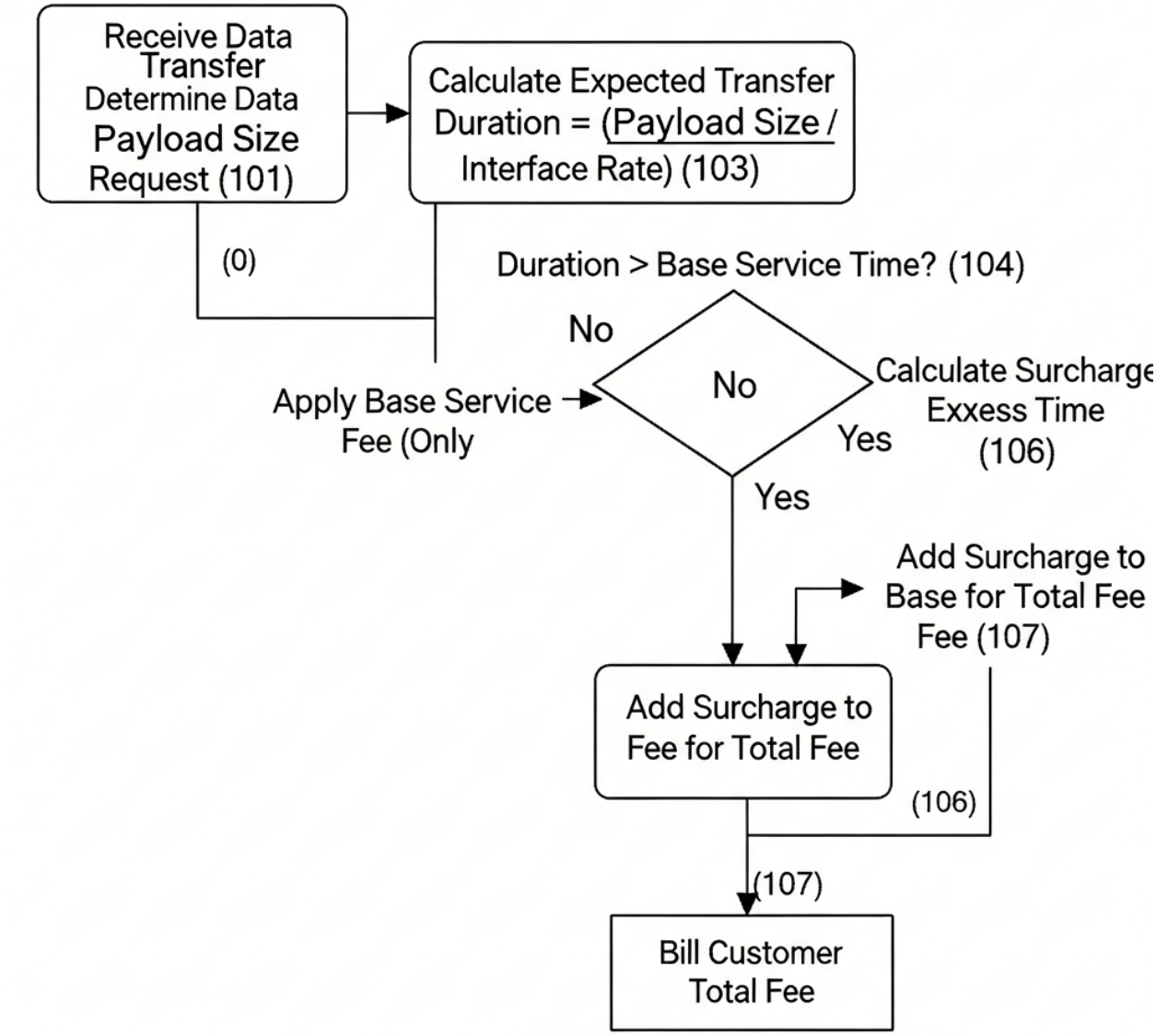


Figure 1: A flowchart illustrating the steps of calculating and applying the payload size surcharge.